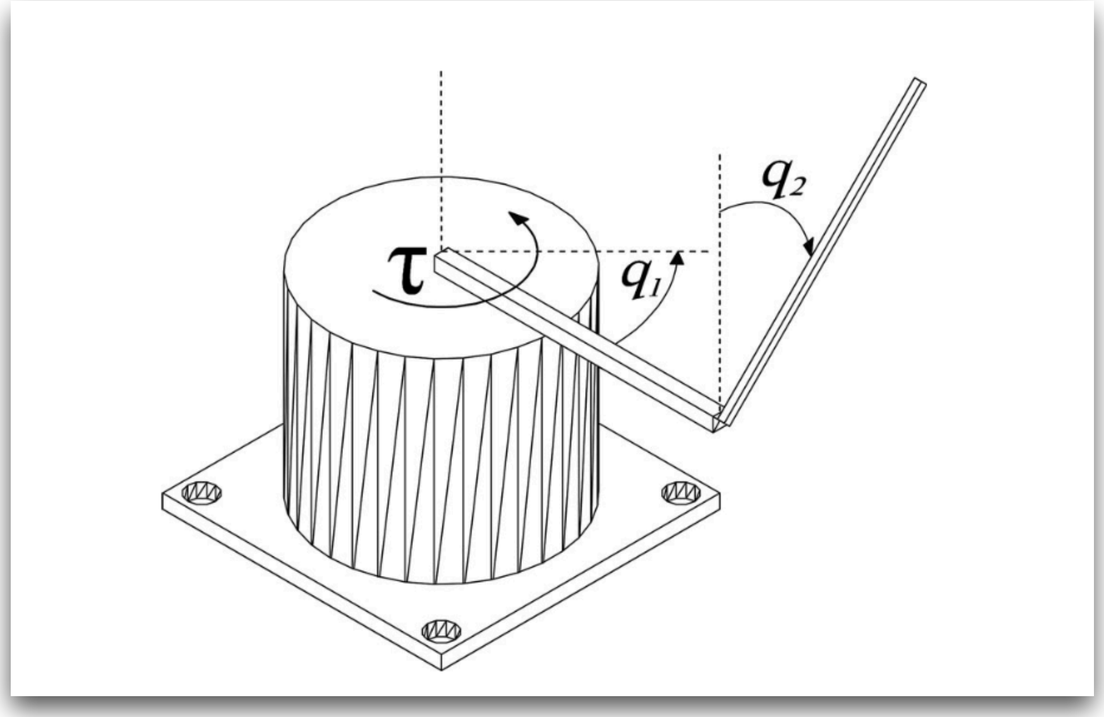
The Furuta Pendulum

Euler–Lagrange Dynamic Modeling of the Furuta Pendulum:

$$M(\boldsymbol{q})\ddot{\boldsymbol{q}} + C(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) + \boldsymbol{f}_v(\dot{\boldsymbol{q}}) + \boldsymbol{f}_c(\dot{\boldsymbol{q}}) = \boldsymbol{u}$$



M(q) : Cumulative inertia matrix

C(q) : Coriolis torque vector

g(q) : Gravitational torque vector

f_v ($\dot{q}$) : Viscous friction vector

f_c ($\dot{q}$) : Coulomb friction vector

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$$M(\boldsymbol{q}) = \begin{bmatrix} \theta_1 + \theta_2 \sin^2(q_2) & \theta_3 \cos(q_2) \\ \theta_3 \cos(q_2) & \theta_4 \end{bmatrix}$$

$$C(\boldsymbol{q}, \dot{\boldsymbol{q}}) =$$

$$\begin{bmatrix} \frac{1}{2}\theta_2 \dot{q}_2 \sin(2q_2) & -\theta_3 \dot{q}_2 \sin(q_2) + \frac{1}{2}\theta_2 \dot{q}_1 \sin(2q_2) \\ -\frac{1}{2}\theta_2 \dot{q}_1 \sin(2q_2) & 0 \end{bmatrix}$$

$$\boldsymbol{g}(\boldsymbol{q}) = \begin{bmatrix} 0 \\ -\theta_5 \sin(q_2) \end{bmatrix}, \; \boldsymbol{f}_v(\dot{\boldsymbol{q}}) = \begin{bmatrix} \theta_6 \dot{q}_1 \\ \theta_7 \dot{q}_2 \end{bmatrix}$$

$$\boldsymbol{f}_c(\dot{\boldsymbol{q}}) = \begin{bmatrix} \theta_8 \tanh(\beta \dot{q}_1) \\ \theta_9 \tanh(\beta \dot{q}_2) \end{bmatrix}$$

The Furuta Pendulum

Reformulation of the Open-Loop Dynamics:

$$\frac{d}{dt}\dot{\boldsymbol{q}} = \boldsymbol{f}_z(\boldsymbol{q}, \dot{\boldsymbol{q}}) + \boldsymbol{g}_z(\boldsymbol{q})\tau$$

$$\boldsymbol{f}_z(\boldsymbol{q}, \dot{\boldsymbol{q}}) = M(\boldsymbol{q})^{-1}[-C(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} - \boldsymbol{g}(\boldsymbol{q}) - \boldsymbol{f}_v(\dot{\boldsymbol{q}}) - \boldsymbol{f}_c(\dot{\boldsymbol{q}})]$$

$$\boldsymbol{g}_z(\boldsymbol{q}) = \frac{1}{\det M}\begin{bmatrix} M_{22} \\ -M_{12} \end{bmatrix}$$

Tracking error vector :

$$\boldsymbol{e} = \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} q_{d1} - q_1 \\ q_{d2} - q_2 \end{bmatrix}$$

q_{d1} (t): Desired trajectory for arm position

q_{d2} (t): Desired pendulum position, selected as zero

The Furuta Pendulum

open-loop error dynamics :

$$\frac{d}{dt}\boldsymbol{x} = \boldsymbol{f}(t, \boldsymbol{x}) + \boldsymbol{g}(\boldsymbol{x})\tau$$

where

$$\boldsymbol{f}(t, \boldsymbol{x}) = \begin{bmatrix} \dot{e}_1 \\ \dot{e}_2 \\ \ddot{q}_{d1} - f_{z1} \\ -f_{z2} \end{bmatrix}, \ \boldsymbol{g}(\boldsymbol{x}) = \begin{bmatrix} 0 \\ 0 \\ -g_{z1} \\ -g_{z2} \end{bmatrix}, \text{ and } \boldsymbol{x} = \begin{bmatrix} e_1 \\ e_2 \\ \dot{e}_1 \\ \dot{e}_2 \end{bmatrix}$$

Control problem :

$$\|\boldsymbol{x}(t_0)\| \leq a \ \Rightarrow \ \|\boldsymbol{x}(t)\| \leq b \ \ \forall\, t \geq t_0 + T$$

with $T = T(a, b) \geq 0$.

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output of the system: $$y = \Delta_1 e_1 + \Delta_2 e_2 + \dot{e}_1 + \dot{e}_2$$

The time derivative of the output function y(t) :

$$\frac{d}{dt}y = \ddot{q_{d1}} - f_{z1} - f_{z2} + \Delta_1 \dot{e}_1 + \Delta_2 \dot{e}_2 - [g_{z1} + g_{z2}]\tau$$

Linearizing Control Input : $$\tau = \frac{-\Delta_1 \dot{e}_1 - \Delta_2 \dot{e}_2 - \ddot{q}_{d1} + f_{z1} + f_{z2} - K_p y}{-[g_{z1} + g_{z2}]}$$

Closed-Loop Dynamics for Output y with Relative Degree One : $$\frac{d}{dt}y = -K_p y$$

$$\lim_{t \to \infty} y(t) = 0$$

The Furuta Pendulum

Simulation Parameters of the Feedback Linearization-Based Control :

Symbol	Value	Unit
θ_1	0.0619	$\mathrm{Kg \cdot m^2 \cdot rad}$
θ_2	0.0149	$\mathrm{Kg \cdot m^2 \cdot rad}$
θ_3	0.0185	$\mathrm{Kg \cdot m^2 \cdot rad}$
θ_4	0.0131	$\mathrm{Kg \cdot m^2 \cdot rad}$
θ_5	0.5076	$\mathrm{Kg \cdot m^2 \cdot rad}$
θ_6	0.0083	$\mathrm{N \cdot m \cdot rad/s}$
θ_7	0.0007	$\mathrm{N \cdot m \cdot rad/s}$
θ_8	0.0188	$\mathrm{N \cdot m \cdot rad/s}$
θ_9	0.0087	$\mathrm{N \cdot m \cdot rad/s}$

$$q_{d1}(t) = \sin(\omega t)$$

$$\beta = 100$$

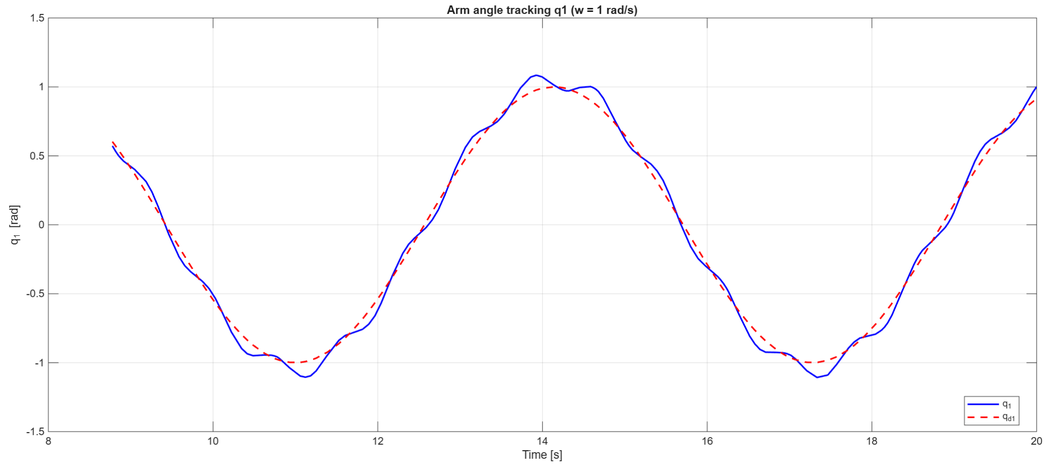
$$\omega = 1 \text{ [rad/s]}$$

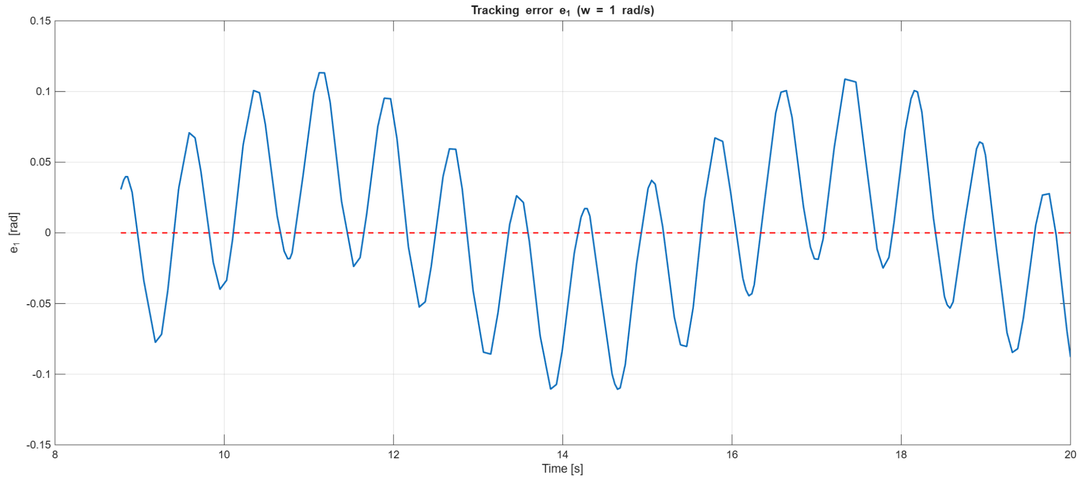
$$\omega = 2.5 \text{ [rad/s]}$$

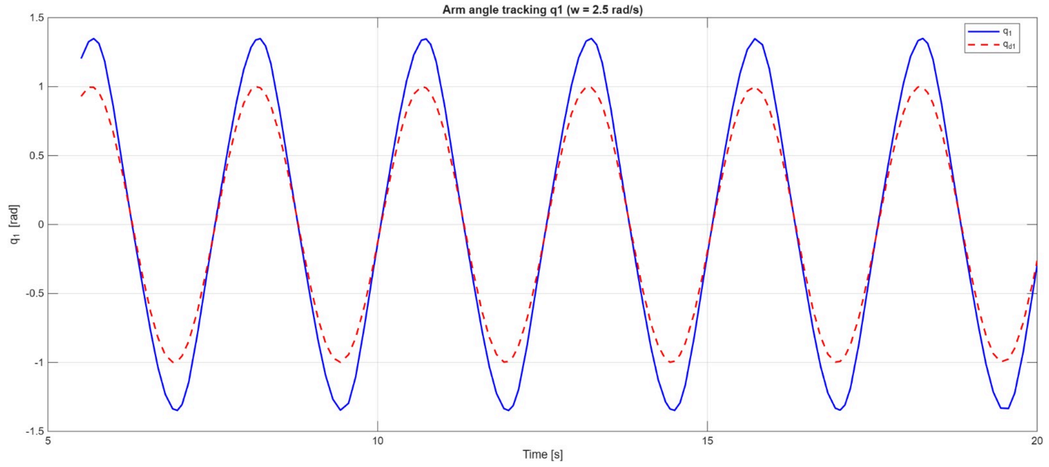
$$K_p = 2,\ \Delta_1 = 6,\ \Delta_2 = 8$$

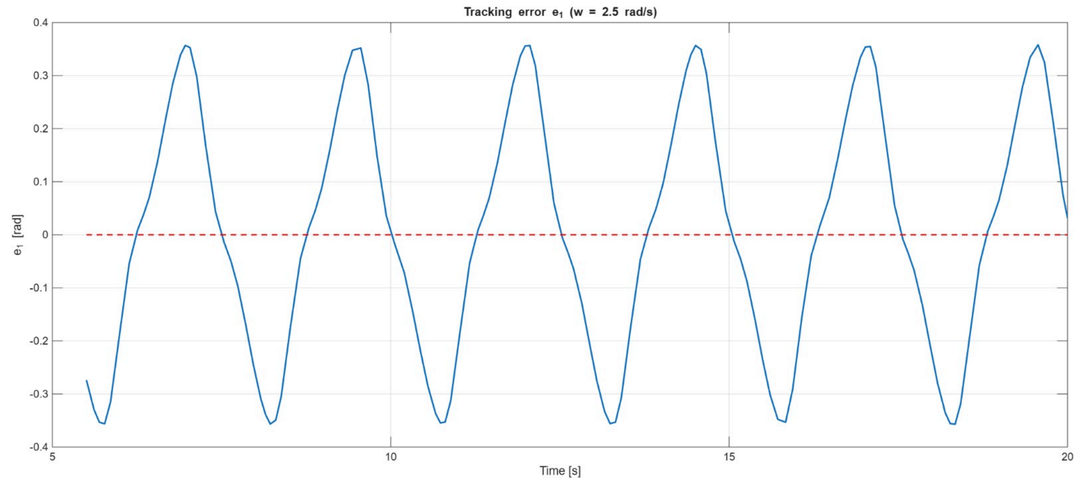
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Experimental Results:



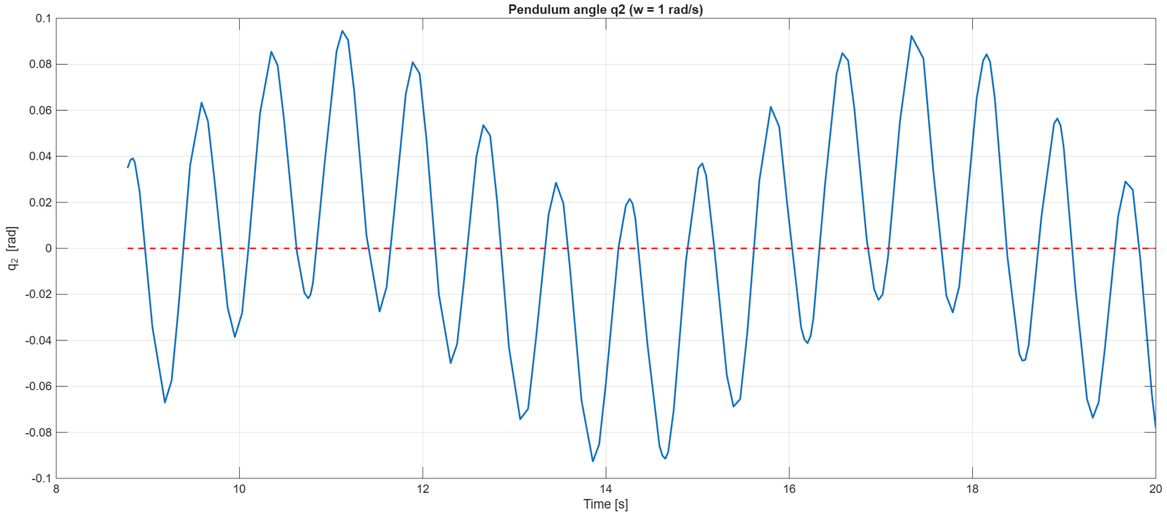


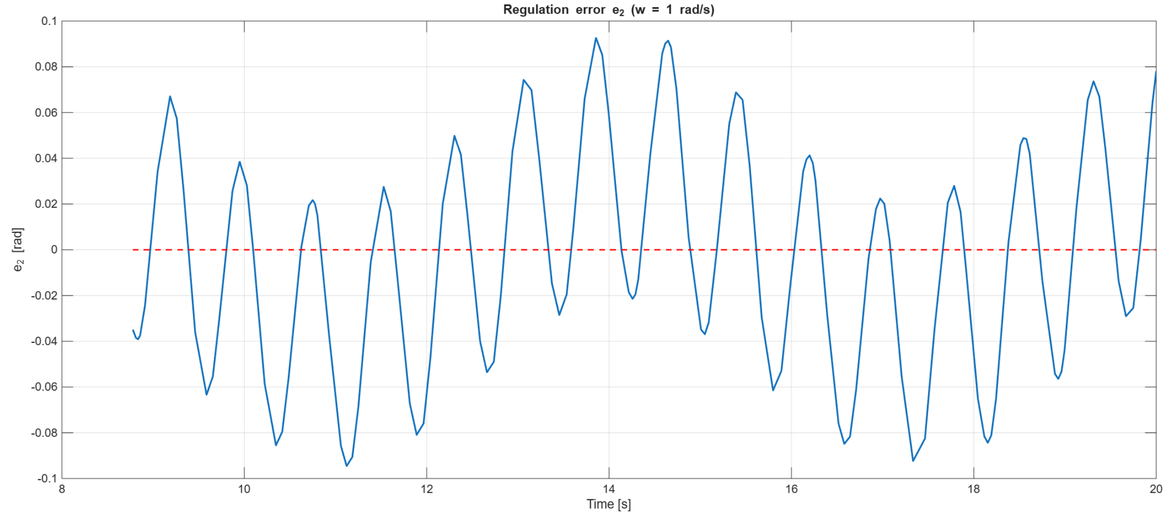


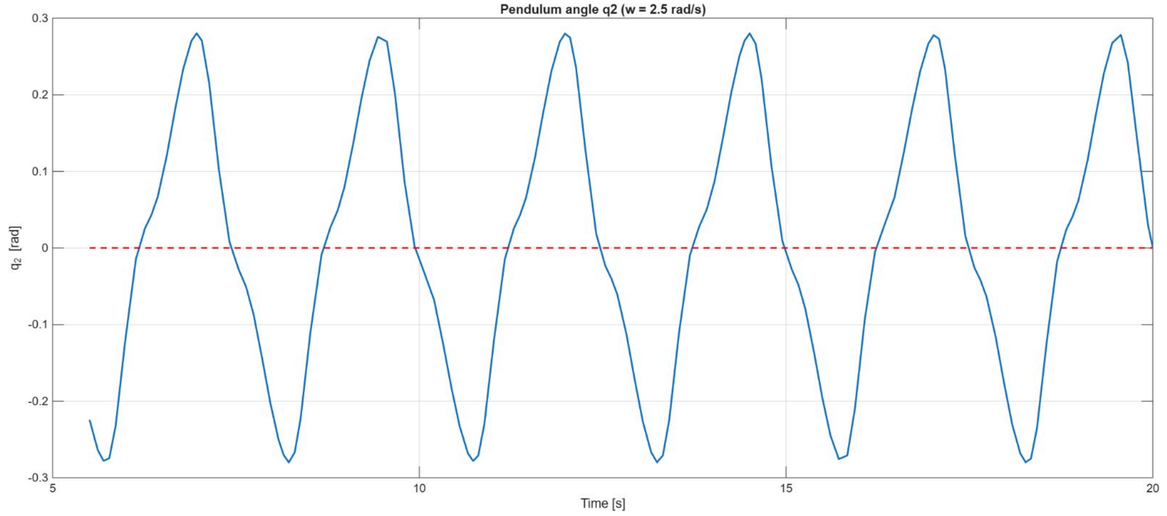


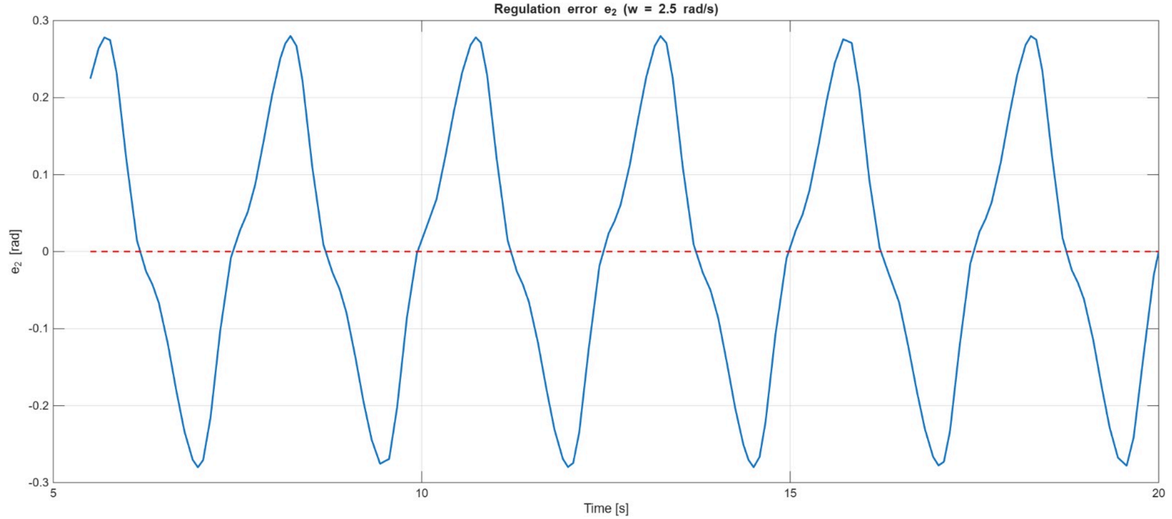
The Furuta Pendulum

Experimental Results:



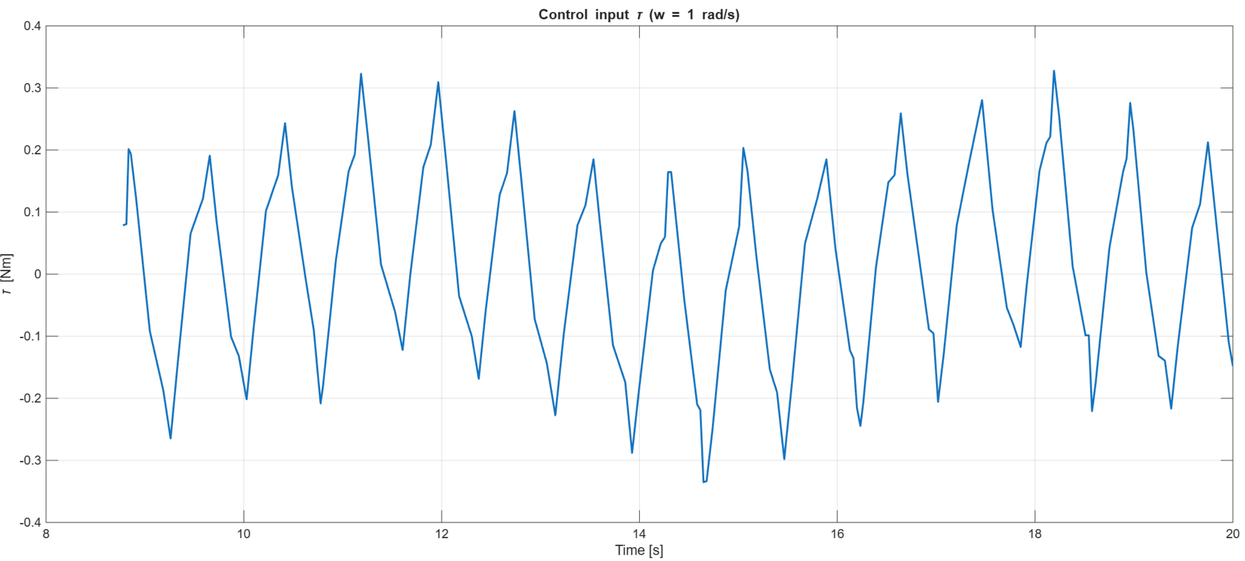


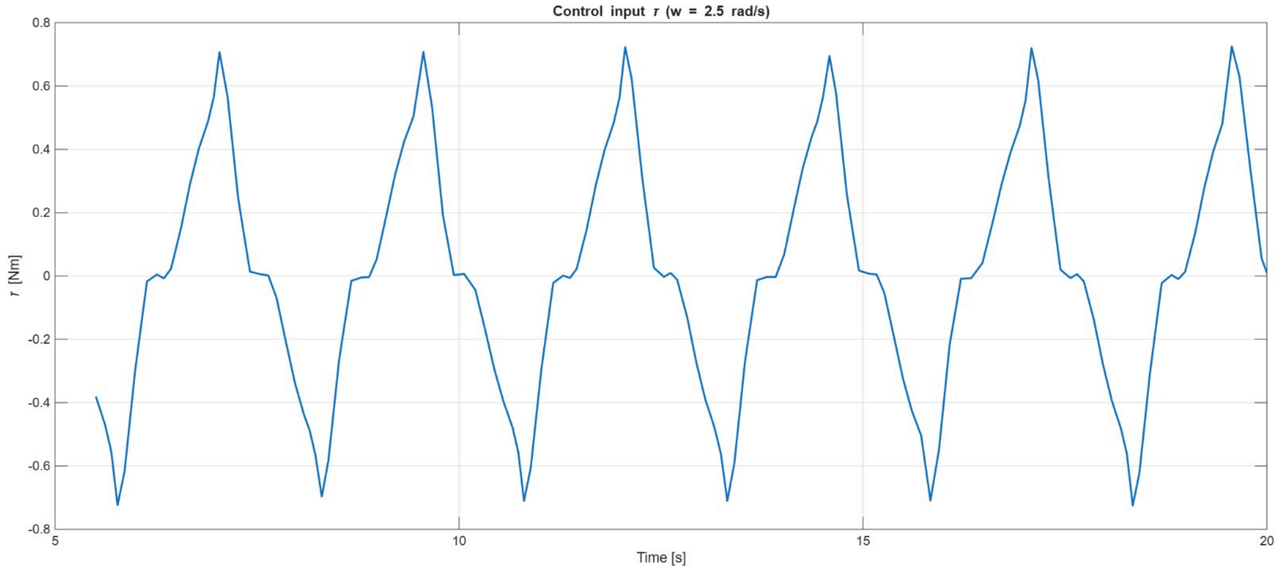




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Experimental Results:

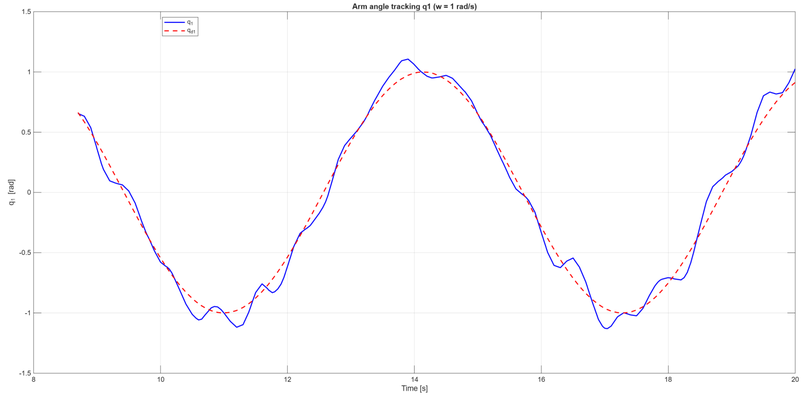


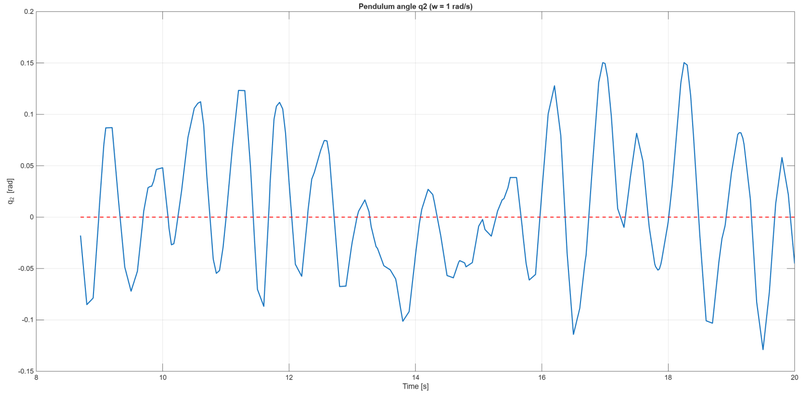


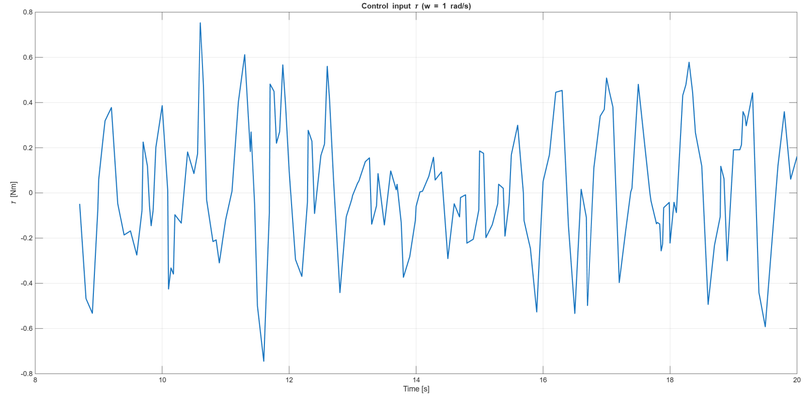
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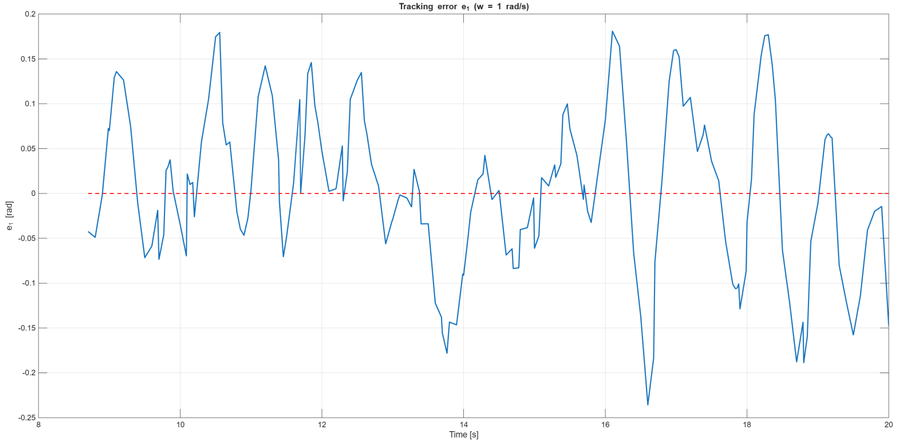
Questions :

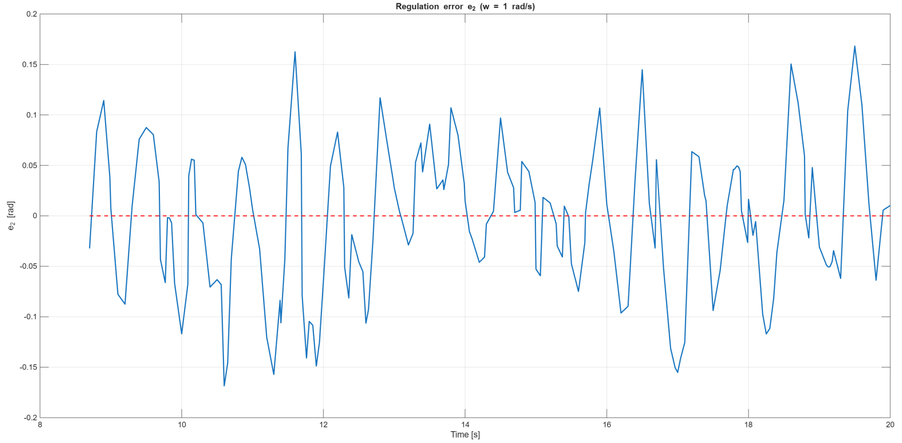
- Add a small output measurement noise and reprot the results. How can you increase the robustness of the controller against output noise?







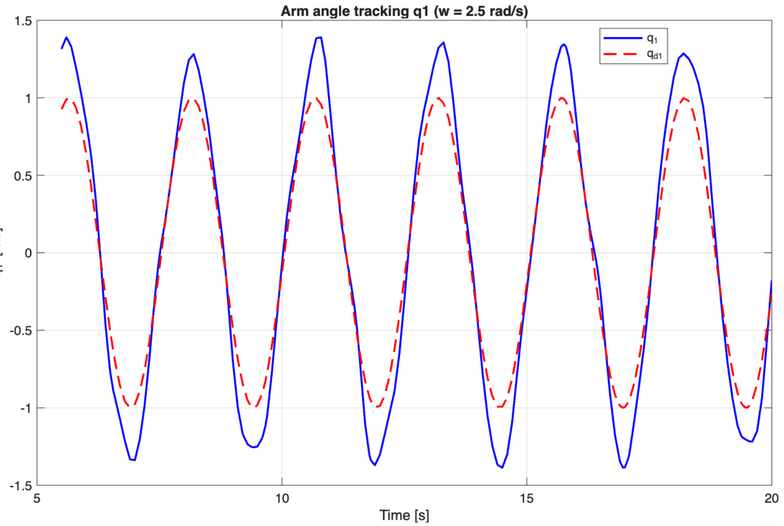


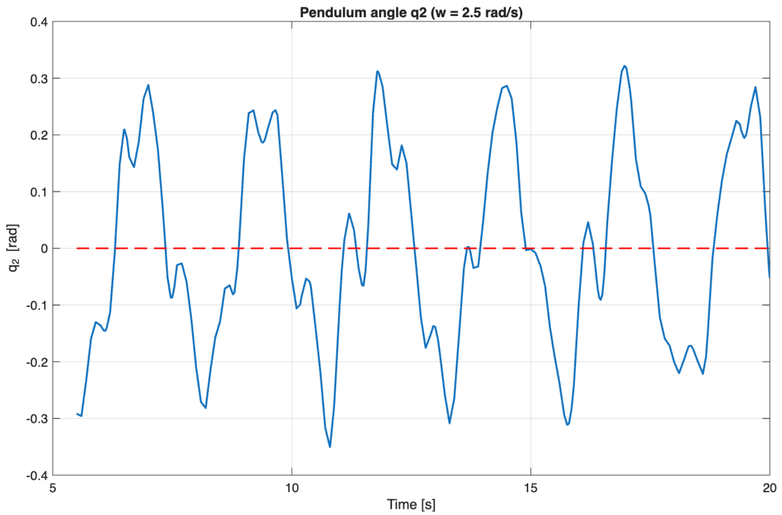


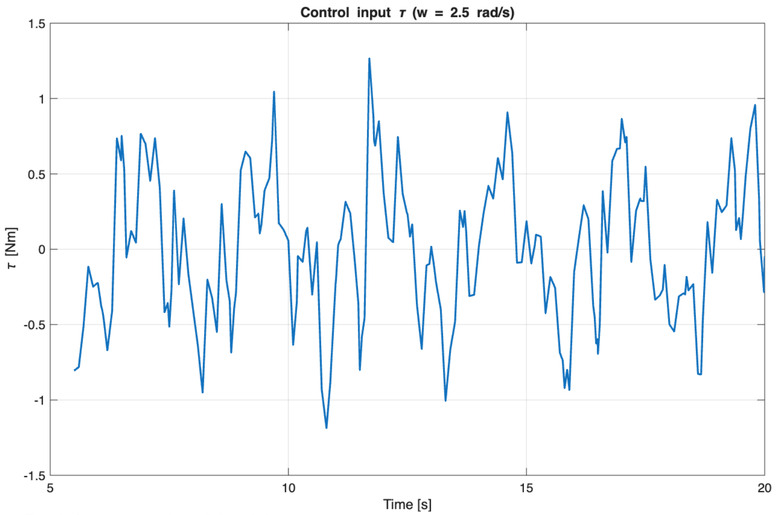
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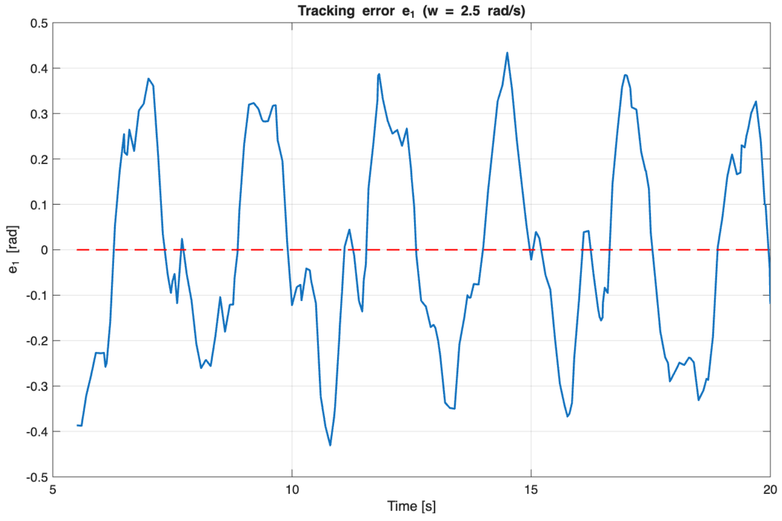
Questions :

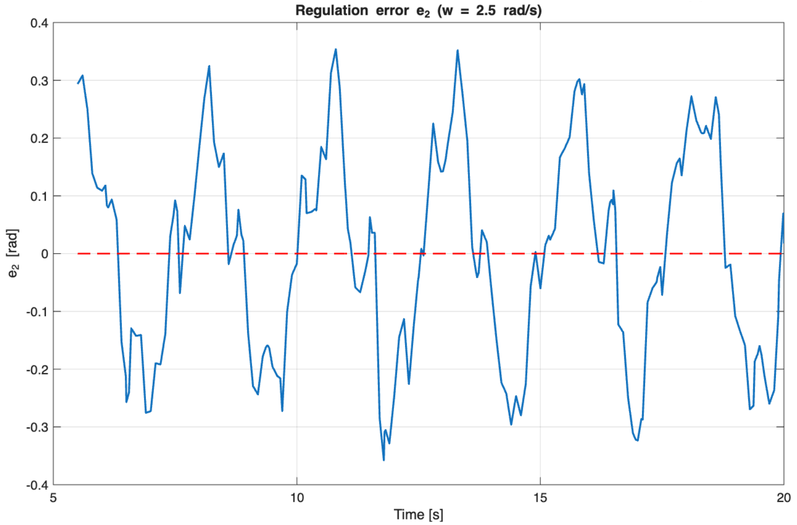
- Add a small output measurement noise and reprot the results. How can you increase the robustness of the controller against output noise?











The Furuta Pendulum

Questions :

- In the paper, the authors are concerned with arm trajectory tracking problem. Initialize the simulation from different initial conditions and report your results for a swing-up control problem.

